5 The matrix **M** represents a sequence of two transformations in the x-y plane given by a one-way stretch in the x-direction, scale factor 3, followed by a reflection in the line $y = x$.

(a) Find **M**. [3]

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(b) Give full details of the geometrical transformation in the x-y plane represented by $\mathbf{M}^{-1}$. [3]

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The matrix **N** is such that $\mathbf{MN} = \begin{pmatrix} 1 & 2 \\ 3 & 2 \end{pmatrix}$.

(c) Find **N**. [3]

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(d) Find the equations of the invariant lines, through the origin, of the transformation in the x-y plane represented by **MN**. [5]